

Generalized Vibrational Analysis

Part I: Linear Algebra on a Metric Space

Goal: establish vectors, covectors, the metric, and projection operators within a general metric space.

Definition 1. Vectors and covectors. Let V be an n -dimensional real vector space with basis $\{e_i\}$. Every vector $v \in V$ has components v^i with $v = v^i e_i$, where repeated upper and lower indices are summed. A *covector* is a linear map $\alpha : V \rightarrow \mathbb{R}$, and the covectors form the *dual space* V^* . The components of a covector are $\alpha_i \equiv \alpha(e_i)$, so that $\alpha(v) = \alpha_i v^i$.

Definition 2. Metric. A *metric* on V is a symmetric, positive-definite bilinear form $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$. Its components are $G_{ij} \equiv \langle e_i, e_j \rangle = G_{ji}$, so that

$$\langle v, w \rangle = G_{ij} v^i w^j.$$

Definition 3. Lowering. The *lowering* (flat) of a vector v is the covector $v^\flat \equiv \langle v, \cdot \rangle$, i.e. $v^\flat(w) = \langle v, w \rangle$. Its components are written $v_i \equiv (v^\flat)_i$.

Proposition 1. Lowering in components.¹

$$v_i = G_{ij} v^j \quad \langle v, w \rangle = v_i w^i$$

The inner product is the contraction of a vector with the lowered form of the other.

Proposition 2. Lowering is invertible.² The lowering operation, $\flat : V \rightarrow V^*$, is a linear and invertible mapping.

Definition 4. Raising. The *raising* (sharp) map $\sharp : V^* \rightarrow V$ is the inverse of $\flat$. Given a vector v which maps to a co-vector $\alpha = v^\flat$ upon lowering, raising maps the co-vector α back to the vector, $\alpha^\sharp = v$. Its components are written $\alpha^i \equiv (\alpha^\sharp)^i$.

Definition 5. Contravariant metric. The raised components of a covector are $\alpha^i \equiv (\alpha^\sharp)^i$. Since $\sharp$ is linear (as the inverse of the linear map $\flat$, Proposition 2), they are linear in the components α_j , and the *contravariant metric* G^{ij} is the matrix relating them:

$$\alpha^i = G^{ij} \alpha_j.$$

Lemma 1. Co- and contravariant metrics.³ The co- and contravariant metrics are matrix inverses, and G^{ij} is symmetric.

$$G^{ik} G_{kj} = G_{jk} G^{ki} = \delta_j^i \quad G^{ij} = G^{ji}$$

¹*Proof:* By Definition 1 and Definition 3, we have $\langle v, w \rangle = v^\flat(w) = v^\flat(e_i) w^i = v_i w^i$, which proves the second result. Comparing this with Definition 2 implies the first result.

²*Proof:* Linearity follows from bilinearity of the inner product: given an arbitrary linear combination, $v = c^I v_I$, the inner product satisfies $\langle v, w \rangle = c^I \langle v_I, w \rangle$, which implies $v^\flat = c^I v_I^\flat$. To prove invertibility, first note that *only* the zero vector maps to the zero-covector: $v^\flat = 0 \Rightarrow \langle v, v \rangle = v^\flat(v) = 0 \Rightarrow v = 0$. It therefore follows that the mapping is one-to-one: $v^\flat = w^\flat \Rightarrow v^\flat - w^\flat = (v - w)^\flat = 0 \Rightarrow v - w = 0 \Rightarrow v = w$. This proves the result, because a one-to-one linear mapping between spaces of equal dimension is automatically invertible.

³*Proof:* For every covector, Definition 4, Definition 5, and Proposition 1 give $\alpha_i = \left((\alpha^\sharp)^\flat \right)_i = G_{ij} \alpha^j = G_{ij} G^{jk} \alpha_k$. Since α is arbitrary, $G_{ij} G^{jk} = \delta_i^k$, so G^{ij} is the matrix inverse of G_{ij} , which is also its left inverse. The inverse of a symmetric matrix is symmetric.

Remark 1. Dual consistency. By construction, $(v^b)^\sharp = v$ and $(\alpha^\sharp)^b = \alpha$. The position of an index therefore tells us which object a component belongs to, and we write v_i and α^i freely.

Definition 6. Covector inner product. The inner product of covectors is $\langle \alpha, \beta \rangle \equiv \langle \alpha^\sharp, \beta^\sharp \rangle$.

Proposition 3. Covector inner product in components.⁴

$$\langle \alpha, \beta \rangle = G^{ij} \alpha_i \beta_j = \alpha_i \beta^i \quad \langle v^b, w^b \rangle = \langle v, w \rangle$$

Definition 7. Orthogonal projector. An *orthogonal projector* is a linear map $P : V \rightarrow V$, with components $(Pv)^i = P_j^i v^j$, that is idempotent and self-adjoint:

$$P^2 = P \quad \langle Pv, w \rangle = \langle v, Pw \rangle.$$

Its complement is $Q = 1 - P$.

Theorem 1. Projection onto a vector.⁵ For $v \neq 0$, the orthogonal projector onto $\text{span}\{v\}$ is

$$P_v = \frac{v \otimes v^b}{\langle v, v \rangle} \quad (P_v)^i_j = \frac{v^i v_j}{\langle v, v \rangle} \quad P_v w = v \frac{\langle v, w \rangle}{\langle v, v \rangle},$$

and $Q_v = 1 - P_v$ is the orthogonal projector onto the orthogonal complement of v .

Corollary 1. Projection of covectors.⁶ An orthogonal projector P acts on covectors from the right, $(\alpha P)_j \equiv \alpha_i P_j^i$ (i.e. $\alpha P = \alpha \circ P$), and this is an orthogonal projector on V^* with respect to the covector inner product, satisfying $(Pv)^b = v^b P$. In particular, for $\alpha \neq 0$, the projector onto $\text{span}\{\alpha\}$ is

$$(P_\alpha)^i_j = \frac{\alpha^i \alpha_j}{\langle \alpha, \alpha \rangle} \quad \beta P_\alpha = \alpha \frac{\langle \alpha, \beta \rangle}{\langle \alpha, \alpha \rangle}.$$

Definition 8. Projection of a bilinear form. The projection of a bilinear form H_{ij} by an orthogonal projector P is $H(Pv, Pw)$, with components

$$(H^P)_{ij} \equiv P_i^k H_{kl} P_j^l.$$

Linear maps between two spaces

From here on, Greek indices μ, ν, ρ, σ refer to a space V of dimension n and Latin indices i, j, k, l refer to a space W of dimension $m \leq n$.

Definition 9. Linear map and pullback. A linear map $L : V \rightarrow W$ has components $(Lv)^i = L_\mu^i v^\mu$. Its pullback (transpose) $L^* : W^* \rightarrow V^*$ is $L^* \beta \equiv \beta \circ L$, with components $(L^* \beta)_\mu = \beta_i L_\mu^i$.

⁴Proof: By Definition 6, Definition 3, Definition 4, and Definition 5, $\langle \alpha, \beta \rangle = \langle \alpha^\sharp, \beta^\sharp \rangle = (\alpha^\sharp)^\flat (\beta^\sharp)^\flat = \alpha (\beta^\sharp)^\flat = \alpha_i \beta^i = G^{ij} \alpha_i \beta_j$. The second identity follows from Remark 1: $\langle v^b, w^b \rangle = \langle (v^b)^\sharp, (w^b)^\sharp \rangle = \langle v, w \rangle$.

⁵Proof: Let $N = \langle v, v \rangle > 0$. By Proposition 1, $(P_v w)^i = v^i v_j w^j / N = v^i \langle v, w \rangle / N$, which is the third form. Since $P_v v = v$, the range is $\text{span}\{v\}$ and $P_v^2 w = P_v v \langle v, w \rangle / N = P_v w$. It is self-adjoint because $\langle P_v w, u \rangle = \langle v, w \rangle \langle v, u \rangle / N$ is symmetric in w and u . For the complement, $Q_v^2 = 1 - 2P_v + P_v^2 = Q_v$ and Q_v is self-adjoint. Its range is orthogonal to v , since $\langle v, Q_v w \rangle = \langle v, w \rangle - \langle v, w \rangle = 0$, and $Q_v w = w$ whenever $\langle v, w \rangle = 0$.

⁶Proof: Comparing $\langle Pv, w \rangle = G_{ki} P_l^i v^l w^k$ with $\langle v, Pw \rangle = G_{li} v^l P_k^i w^k$, self-adjointness reads $G_{ki} P_l^i = G_{li} P_k^i$. Hence $((Pv)^b)_j = G_{ji} P_l^i v^l = G_{li} v^l P_j^i = v_i P_j^i = (v^b P)_j$. Idempotence is immediate: $\alpha P P = \alpha P^2 = \alpha P$. Taking $v = \alpha^\sharp$ in the identity above gives $(\alpha P)^\sharp = P \alpha^\sharp$, so $\langle \alpha P, \beta \rangle = \langle P \alpha^\sharp, \beta^\sharp \rangle = \langle \alpha^\sharp, P \beta^\sharp \rangle = \langle \alpha, \beta P \rangle$. Finally, $P_\alpha \equiv P_{\alpha^\sharp}$ from Theorem 1 has components $\alpha^i \alpha_j / \langle \alpha, \alpha \rangle$ by Remark 1 and Definition 6, and $\beta_i \alpha^i = \langle \alpha, \beta \rangle$ by Proposition 3.

Proposition 4. Pullback of a surjection is injective.⁷ If L is surjective, then L^* is injective.

Definition 10. Induced metric. Given a metric on V and a surjective $L : V \rightarrow W$, the *induced* covector inner product on W^* is

$$\langle \beta, \gamma \rangle_W \equiv \langle L^* \beta, L^* \gamma \rangle_V.$$

The induced metric on W is the one whose raising and lowering reproduce this covector inner product.

Proposition 5. Induced metric in components.⁸ The induced contravariant metric

$$G^{ij} = L_\mu^i L_\nu^j G^{\mu\nu}$$

is symmetric and positive-definite. Its inverse G_{ij} is therefore the induced (covariant) metric on W .

Theorem 2. Minimum-norm lift.⁹ Let $L : V \rightarrow W$ be surjective and equip W with the induced metric. Define $A : W \rightarrow V$ by $A \equiv \sharp_V \circ L^* \circ \flat_W$, i.e.

$$A_i^\mu = G^{\mu\nu} L_\nu^k G_{ki}.$$

Then: (a) $LA = 1_W$, i.e. $L_\mu^i A_j^\mu = \delta_j^i$; (b) AL is the orthogonal projector on V onto $(\ker L)^\perp$; (c) Aw is the unique vector of minimum norm with $Lv = w$; (d) $\langle Aw, Aw' \rangle_V = \langle w, w' \rangle_W$.

Remark 2. Interpretation of the lift. A is simply L with both of its indices moved by the metrics. When $m = n$, $A = L^{-1}$ and $AL = 1_V$.

⁷ Proof: If $L^* \beta = 0$, then $\beta(Lv) = 0$ for all $v \in V$. Since L is surjective, $\beta(w) = 0$ for all $w \in W$, so $\beta = 0$.

⁸ Proof: By Definition 9 and Proposition 3, $\langle \beta, \gamma \rangle_W = G^{\mu\nu} (\beta_i L_\mu^i) (\gamma_j L_\nu^j) = G^{ij} \beta_i \gamma_j$. Symmetry follows from that of $G^{\mu\nu}$. For $\beta \neq 0$, $L^* \beta \neq 0$ by Proposition 4, so $G^{ij} \beta_i \beta_j = \langle L^* \beta, L^* \beta \rangle_V > 0$. A symmetric positive-definite matrix has a symmetric positive-definite inverse G_{ij} , which is therefore a metric on W . By Lemma 1 its raising matrix is G^{ij} , so by Proposition 3 it reproduces the induced covector inner product.

⁹ Proof: First, for any $v, u \in V$, $\langle ALv, u \rangle_V = G_{\mu\rho} G^{\mu\nu} L_\nu^k G_{ki} L_\sigma^i v^\sigma u^\rho = G_{ki} (Lu)^k (Lv)^i = \langle Lv, Lu \rangle_W$. (†) (a) $L_\mu^i A_j^\mu = L_\mu^i L_\nu^k G^{\mu\nu} G_{kj} = G^{ik} G_{kj} = \delta_j^i$ by Proposition 5. (b) Let $P = AL$. By (a), $P^2 = A(LA)L = P$, and by (†), P is self-adjoint. Also by (a), $LP = L$, so $Pv = 0$ if and only if $Lv = 0$, i.e. $\ker P = \ker L$. The range of a self-adjoint projector is the orthogonal complement of its kernel (if $Pu = 0$, then $\langle Pv, u \rangle = \langle v, Pu \rangle = 0$, and the dimensions add to n). (c) $LAw = w$ by (a). If $Lv = w$, then $Pv = Aw$, so $v = Aw + (1 - P)v$ is an orthogonal decomposition by (b) and $\langle v, v \rangle_V = \langle Aw, Aw \rangle_V + \langle (1 - P)v, (1 - P)v \rangle_V$, which is minimized only by $v = Aw$. (d) By (a) and (†), $\langle Aw, Aw' \rangle_V = \langle ALAw, Aw' \rangle_V = \langle LAw, LAw' \rangle_W = \langle w, w' \rangle_W$.

Part II: Coordinates and Derivatives

Goal: show that coordinate displacements are vectors and derivatives are covectors, so that the machinery of Part I applies to them.

Definition 11. Displacements and differentials. Let q^μ be coordinates on a neighborhood of a reference point in configuration space. Infinitesimal displacements $d\mathbf{q}$, with components dq^μ , are vectors. For a smooth function f , the differential $df(d\mathbf{q}) = \frac{\partial f}{\partial q^\mu} dq^\mu$ is linear in $d\mathbf{q}$, so df is a covector with components $\frac{\partial f}{\partial q^\mu}$.

Definition 12. Coordinate map. A second set of $m \leq n$ coordinates $q^i = q^i(q^\mu)$ has *Jacobian* and *second derivative*

$$B_\mu^i \equiv \frac{\partial q^i}{\partial q^\mu} \quad C_{\mu\nu}^i \equiv \frac{\partial^2 q^i}{\partial q^\mu \partial q^\nu} = C_{\nu\mu}^i.$$

We assume B_μ^i has full row rank at the reference point, so that B is surjective.

Proposition 6. Displacements push forward.¹⁰

$$dq^i = B_\mu^i dq^\mu$$

Proposition 7. Gradients pull back.¹¹ If f depends on q^μ only through q^i , then

$$\frac{\partial f}{\partial q^\mu} = \frac{\partial f}{\partial q^i} B_\mu^i,$$

i.e. df in the q^μ coordinates is the pullback B^* of df in the q^i coordinates.

Remark 3. Transformation laws. When $m = n$ and B is invertible, these become the classical contravariant and covariant transformation laws, and the contraction $\frac{\partial f}{\partial q^i} dq^i = \frac{\partial f}{\partial q^\mu} dq^\mu$ is invariant. When $m < n$, displacements can only be pushed forward and gradients can only be pulled back.

Proposition 8. Second derivatives.¹² If f depends on q^μ only through q^i , then

$$\frac{\partial^2 f}{\partial q^\mu \partial q^\nu} = \frac{\partial^2 f}{\partial q^i \partial q^j} B_\mu^i B_\nu^j + \frac{\partial f}{\partial q^k} C_{\mu\nu}^k.$$

Second derivatives therefore transform as a bilinear form only when the map is linear ($C = 0$) or at a stationary point ($\partial f / \partial q^k = 0$).

¹⁰Proof: This is the chain rule applied to $q^i(q^\mu)$: to first order, $dq^i = \frac{\partial q^i}{\partial q^\mu} dq^\mu$.

¹¹Proof: This is the chain rule applied to $f(q^i(q^\mu))$. The components of the pullback are given in Definition 9.

¹²Proof: Differentiate Proposition 7 with respect to q^ν , using the product rule and the chain rule: $\frac{\partial^2 f}{\partial q^i \partial q^j} B_\nu^j B_\mu^i + \frac{\partial f}{\partial q^k} \frac{\partial B_\mu^k}{\partial q^\nu}$, where $\partial B_\mu^k / \partial q^\nu = C_{\mu\nu}^k$.

Part III: The Physical Model

Goal: identify the kinetic energy as the metric, and apply Part I to Cartesian, mass-weighted Cartesian, and internal coordinates.

Definition 13. Cartesian coordinates and kinetic-energy metric. For N atoms with Cartesian coordinates x^μ ($\mu = 1, \dots, 3N$), let $m_{(\mu)}$ denote the mass of the atom to which coordinate μ belongs. Cartesian displacements from a reference geometry are $q^\mu = x^\mu - x_0^\mu$. The *kinetic-energy metric* is

$$G_{\mu\nu} \equiv m_{(\mu)} \delta_{\mu\nu},$$

chosen so that $T = \frac{1}{2} \langle \dot{q}, \dot{q} \rangle$.

Proposition 9. Contravariant Cartesian metric.¹³

$$G^{\mu\nu} = m_{(\mu)}^{-1} \delta^{\mu\nu}$$

Definition 14. Mass-weighted coordinates. The mass-weighted Cartesian displacements are $\tilde{q}^\mu \equiv m_{(\mu)}^{\frac{1}{2}} q^\mu$. Mass-weighted quantities are marked with a tilde.

Proposition 10. Mass-weighted metric.¹⁴

$$\tilde{G}_{\mu\nu} = \delta_{\mu\nu} \quad \tilde{G}^{\mu\nu} = \delta^{\mu\nu}$$

Definition 15. Internal coordinates. Internal coordinates q^i ($i = 1, \dots, M$, with $M = 3N - 6$, or $3N - 5$ for linear molecules) are displacements of non-redundant functions of the geometry that are invariant to overall translation and rotation. Their mass-weighted Jacobian (B-matrix) and its derivative (C-tensor) are

$$\tilde{B}_\mu^i \equiv \frac{\partial q^i}{\partial \tilde{q}^\mu} \quad \tilde{C}_{\mu\nu}^i \equiv \frac{\partial^2 q^i}{\partial \tilde{q}^\mu \partial \tilde{q}^\nu},$$

and $\tilde{B}_\mu^i$ has full row rank at the reference geometry. Index-free internal-coordinate objects are marked with a prime, e.g. v' .

Theorem 3. Wilson G-matrix.¹⁵ The metric induced on internal coordinates by $\tilde{B}$ has contravariant components

$$G^{ij} = \delta^{\mu\nu} \tilde{B}_\mu^i \tilde{B}_\nu^j = \sum_\mu m_{(\mu)}^{-1} B_\mu^i B_\mu^j,$$

where $B_\mu^i = \partial q^i / \partial q^\mu$. The covariant internal-coordinate metric G_{ij} is its inverse.

¹³Proof: By Lemma 1, $G^{\mu\nu}$ is the inverse of the diagonal matrix $m_{(\mu)} \delta_{\mu\nu}$.

¹⁴Proof: The map $q^\mu \rightarrow \tilde{q}^\mu$ is linear and invertible, with Jacobian $m_{(\mu)}^{\frac{1}{2}} \delta_{\mu\nu}^{\frac{1}{2}}$. For an invertible map, the induced metric is the same inner product expressed in the new components (Theorem 2 (d) with $A = L^{-1}$), so Proposition 5 and Proposition 9 give $\tilde{G}^{\mu\nu} = m_{(\mu)}^{\frac{1}{2}} m_{(\nu)}^{\frac{1}{2}} m_{(\mu)}^{-1} \delta^{\mu\nu} = \delta^{\mu\nu}$, whose inverse is $\tilde{G}_{\mu\nu} = \delta_{\mu\nu}$. As a check, $T = \frac{1}{2} \sum_\mu m_{(\mu)} (\dot{q}^\mu)^2 = \frac{1}{2} \sum_\mu (\dot{\tilde{q}}^\mu)^2$.

¹⁵Proof: By Proposition 6, displacements map as $dq^i = \tilde{B}_\mu^i d\tilde{q}^\mu$, and $\tilde{B}$ is surjective (Definition 15). Proposition 5 with $L_\mu^i = \tilde{B}_\mu^i$ and Proposition 10 give the first form. For the second, $q^\nu = m_{(\nu)}^{-\frac{1}{2}} \tilde{q}^\nu$ and Proposition 7 give $\tilde{B}_\mu^i = m_{(\mu)}^{-\frac{1}{2}} B_\mu^i$.

Corollary 2. *A-matrix.*¹⁶ *The A-matrix*

$$\tilde{A}_i^\mu \equiv \delta^{\mu\nu} \tilde{B}_\nu^k G_{ki}$$

satisfies $\tilde{B}_\mu^i \tilde{A}_j^\mu = \delta_j^i$, and $\tilde{A}_i^\mu \tilde{B}_\nu^i$ is the orthogonal projector onto the vibrational subspace $(\ker \tilde{B})^\perp$, where $\ker \tilde{B}$ is spanned by infinitesimal translations and rotations.

Proposition 11. *Kinetic energy in internal coordinates.*¹⁷ *For Cartesian velocities orthogonal to $\ker \tilde{B}$ (no infinitesimal overall translation or rotation, i.e. the linearized Eckart conditions),*

$$T = \frac{1}{2} G_{ij} \dot{q}^i \dot{q}^j.$$

¹⁶*Proof:* Apply Theorem 2 (a) and (b) with $\mathbf{L} = \tilde{\mathbf{B}}$ and $G^{\mu\nu} = \delta^{\mu\nu}$ (Proposition 10). Each q^i is invariant to overall translation and rotation, so its derivative along an infinitesimal translation or rotation vanishes and these lie in $\ker \tilde{\mathbf{B}}$. By rank-nullity, $\dim \ker \tilde{\mathbf{B}} = 3N - M$, which is the number of independent infinitesimal translations and rotations, so they span the kernel.

¹⁷*Proof:* By Proposition 6, $\dot{q}^i = \tilde{B}_\mu^i \dot{\tilde{q}}^\mu$. If $\dot{\tilde{\mathbf{q}}} \perp \ker \tilde{\mathbf{B}}$, then $\dot{\tilde{\mathbf{q}}} = \tilde{\mathbf{A}} \tilde{\mathbf{B}} \dot{\tilde{\mathbf{q}}} = \tilde{\mathbf{A}} \dot{\tilde{\mathbf{q}}}'$ by Corollary 2, and by Theorem 2 (d), $T = \frac{1}{2} \langle \dot{\tilde{\mathbf{q}}}, \dot{\tilde{\mathbf{q}}} \rangle = \frac{1}{2} \langle \tilde{\mathbf{A}} \dot{\tilde{\mathbf{q}}}', \tilde{\mathbf{A}} \dot{\tilde{\mathbf{q}}}' \rangle = \frac{1}{2} G_{ij} \dot{q}^i \dot{q}^j$.

Part IV: Vibrational Analysis

Goal: transform the gradient and Hessian into internal coordinates, set up the GF eigenproblem, and project the gradient direction out of the Hessian.

Definition 16. Gradient and Hessian. The gradient and Hessian in each coordinate system are defined by the Taylor expansion of the potential energy about the reference geometry:

$$V = V_0 + g_\mu q^\mu + \frac{1}{2} H_{\mu\nu} q^\mu q^\nu + \mathcal{O}(q^3)$$

$$V = V_0 + \tilde{g}_\mu \tilde{q}^\mu + \frac{1}{2} \tilde{H}_{\mu\nu} \tilde{q}^\mu \tilde{q}^\nu + \mathcal{O}(\tilde{q}^3)$$

$$V = V_0 + g_i q^i + \frac{1}{2} H_{ij} q^i q^j + \mathcal{O}(q'^3)$$

Proposition 12. Mass-weighted gradient and Hessian.¹⁸

$$\tilde{g}_\mu = m_{(\mu)}^{-\frac{1}{2}} g_\mu \quad \tilde{H}_{\mu\nu} = m_{(\mu)}^{-\frac{1}{2}} m_{(\nu)}^{-\frac{1}{2}} H_{\mu\nu}$$

Theorem 4. Internal-coordinate gradient.¹⁹

$$g_i = \tilde{g}_\mu \tilde{A}_i^\mu$$

Theorem 5. Internal-coordinate Hessian.²⁰

$$H_{ij} = (\tilde{H}_{\mu\nu} - g_k \tilde{C}_{\mu\nu}^k) \tilde{A}_i^\mu \tilde{A}_j^\nu$$

The correction term arises because the map to internal coordinates is non-linear (Proposition 8).

Definition 17. GF matrix. The GF matrix is the internal-coordinate Hessian with its first index raised,

$$(GF)_j^i \equiv G^{ik} H_{kj},$$

a linear map on internal-coordinate displacements. Its eigenvalues are the squared harmonic frequencies ω^2 .

Proposition 13. GF and mass-weighted Cartesian eigenvalues.²¹ Let $\tilde{P}_\nu^\mu = \tilde{A}_i^\mu \tilde{B}_\nu^i$. The eigenvalues of $G^{ik} H_{kj}$ are the eigenvalues of $\tilde{P}_\mu^\rho (\tilde{H}_{\rho\sigma} - g_k \tilde{C}_{\rho\sigma}^k) \tilde{P}_\nu^\sigma$ restricted to the vibrational subspace. At a stationary point, these are the eigenvalues of the projected mass-weighted Hessian.

¹⁸Proof: The Taylor coefficients are the derivatives at the reference geometry, e.g. $g_\mu = \partial V / \partial q^\mu$. The map $\tilde{q}^\mu \rightarrow q^\mu = m_{(\mu)}^{-\frac{1}{2}} \tilde{q}^\mu$ is linear, so Proposition 7 and Proposition 8 (with $C = 0$) apply with Jacobian $m_{(\mu)}^{-\frac{1}{2}} \delta_\nu^\mu$.

¹⁹Proof: The potential is invariant to overall translation and rotation, so near the reference geometry it depends on $\tilde{q}^\mu$ only through the q^i . By Proposition 7, $\tilde{g}_\mu = g_k \tilde{B}_\mu^k$. Contracting with $\tilde{A}_i^\mu$ and using Corollary 2 gives $\tilde{g}_\mu \tilde{A}_i^\mu = g_k \delta_i^k = g_i$.

²⁰Proof: As in Theorem 4, Proposition 8 gives $\tilde{H}_{\mu\nu} - g_k \tilde{C}_{\mu\nu}^k = H_{kl} \tilde{B}_\mu^k \tilde{B}_\nu^l$. Contracting with $\tilde{A}_i^\mu \tilde{A}_j^\nu$ and using Corollary 2 gives the result.

²¹Proof: Let $K_{\mu\nu} = \tilde{H}_{\mu\nu} - g_k \tilde{C}_{\mu\nu}^k = H_{kl} \tilde{B}_\mu^k \tilde{B}_\nu^l$ (proof of Theorem 5). Since $\tilde{B}_\rho^k \tilde{P}_\mu^\rho = \tilde{B}_\mu^k$ by Corollary 2, $\tilde{P}_\mu^\rho K_{\rho\sigma} \tilde{P}_\nu^\sigma = K_{\mu\nu}$; the projectors only make explicit that K vanishes on $\ker \tilde{B}$. Raise the first index of K with $\delta^{\mu\nu}$ to get a linear map $\mathbf{K}$ on V . For any internal u^j , $\delta^{\mu\rho} K_{\rho\nu} \tilde{A}_j^\nu u^j = \delta^{\mu\rho} \tilde{B}_\rho^k H_{kj} u^j$ and $\tilde{A}_i^\mu u^i = \delta^{\mu\rho} \tilde{B}_\rho^k G_{ki} u^i$. Since $\tilde{B}^*$ is injective (Proposition 4), $\mathbf{K} \tilde{A} u' = \lambda \tilde{A} u'$ if and only if $H_{kj} u^j = \lambda G_{ki} u^i$, i.e. $G^{ik} H_{kj} u^j = \lambda u^i$. By Theorem 2, $\tilde{A}$ maps internal displacements bijectively onto $(\ker \tilde{B})^\perp$, so the two eigenproblems are equivalent. At a stationary point, $g_k = 0$ and $K_{\mu\nu} = \tilde{H}_{\mu\nu}$.

Definition 18. Gradient-direction projector. For $g_i \neq 0$, with $g^i = G^{ij}g_j$ and $\langle g', g' \rangle = G^{ij}g_i g_j$, the gradient-direction projector and its complement are

$$(P_g)_j^i = \frac{g^i g_j}{\langle g', g' \rangle} \quad Q_g = 1 - P_g.$$

Theorem 6. Projected Hessian.²² The projected Hessian

$$H_{ij}^Q = (Q_g)_i^k H_{kl} (Q_g)_j^l$$

gives a GF matrix $G^{ik} H_{kj}^Q$ with eigenvector g^i of eigenvalue 0. Every eigenvector v^i with non-zero eigenvalue satisfies $g_i v^i = 0$, and these $M - 1$ eigenvalues are the generalized frequencies ω^2 for motion orthogonal to the gradient.

Remark 4. Comparison with Cartesian projection. To investigate: the relation to Cartesian gradient projection (Miller, Handy, and Adams). Note that $\tilde{A}_i^\mu g^i = \tilde{g}^\mu$, so the gradient directions agree. By Proposition 13, however, the internal-coordinate result carries the curvature term $g_k \tilde{C}_{\mu\nu}^k$, so projected frequencies away from stationary points depend on the choice of coordinates.

²²Proof: Write $Q = Q_g$, so that $Q_j^l = \delta_j^l - g^l g_j / \langle g', g' \rangle$. Then $Q_j^l g^j = g^l - g^l = 0$, so $H_{kj}^Q g^j = 0$ and g^i is an eigenvector of eigenvalue 0. The matrix $G^{ik} H_{kj}^Q$ is self-adjoint with respect to G_{ij} , because $G_{li} G^{ik} H_{kj}^Q = H_{lj}^Q$ is symmetric. It therefore has a G -orthogonal eigenbasis containing g^i , and the remaining $M - 1$ eigenvectors satisfy $G_{ij} g^i v^j = g_j v^j = 0$. Every eigenvector with $\lambda \neq 0$ is of this kind, since $\lambda g_i v^i = g^k H_{kj}^Q v^j = 0$. For such v^i , $Q_j^l v^j = v^l$, so H^Q agrees with H on the hyperplane orthogonal to the gradient.